

CrucibleCam

User's Guide



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Version 1.5 – September 2026

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1. Overview

1.1. Purpose and scope

CrucibleCam is software for defining motion laws, selecting and coupling curves, verifying continuity, generating the geometric profile, and performing the kinematic analysis of radial cams. This guide describes the program's commands and operating criteria. Familiarity with cam terminology, motion laws, followers, and standard design requirements is assumed.

The program divides the cycle into segments. Each segment is assigned an angular interval, a displacement change, and a curve. The resulting solution calculates the profile, produces the SVAJ diagrams, analyzes pressure angle and curvature, and provides the report and technical exports.

1.2. Operational workflow

The recommended sequence separates motion specification from follower geometry definition. Validation should be repeated whenever segments, curves, custom conditions, or follower parameters are changed.

- Define the discretization and input conventions in Settings;
- Enter the segments and choose manual selection or automatic search;
- Review compatibility and continuity before accepting the sequence;
- Define the follower and geometric parameters after obtaining a valid motion law;
- Analyze charts, tables, the profile, pressure angle, and curvature before exporting.

1.3. Interface conventions

Table 1 – Main quantities and symbols

Symbol	Quantity
θ	Cam angle
β	Segment angular span
L	Displacement
S, V, A, J	Displacement, velocity, acceleration, and jerk
V, A / V, A, J	Compatibility criterion

2. CrucibleCam interface

2.1. Main window overview

The main window is divided into three resizable areas. The left column contains the segment definition and cam data; the central area combines the displacement chart with the profile drawing; and the right column contains result charts and tables. The top bar provides the project and analysis tools.

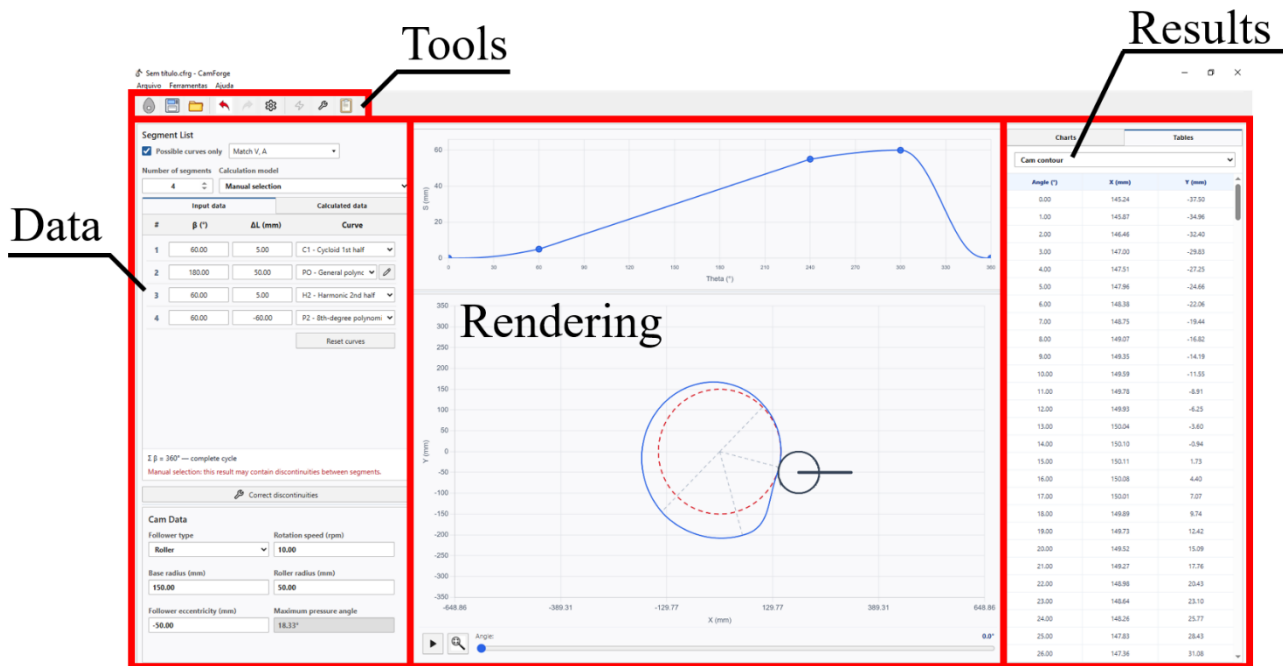


Figure 1 - CrucibleCam main window

The dividers between columns and panels can be dragged. Double-clicking a divider restores its default size. The program retains the selected dimensions for later use.

2.2. Toolbar and menus

Table 2 – Main commands

Command	Function
New project	Creates a new project
Save / Save As	Saves the project to a .ccm file
Open project	Opens a previously saved project
Undo / Redo	Navigates through the project's change history.
Settings	Opens units, resolution, input conventions, and curve options.
Calculate cam	Runs the automatic curve search when this model is selected.
Correct discontinuities	Opens the diagnostics and search for corrections to velocity, acceleration, and, when selected, jerk discontinuities.
Project report	Opens the report preview and allows export to PDF.

2.3. Segment List

The header controls the number of segments, the calculation model, and, in manual selection, the possible-curve filter. The Input Data tab displays the fields defined in Settings. The Calculated Data tab shows the equivalent representation without duplicating the input.

When the number of segments changes, CrucibleCam resizes the list and revalidates the existing curves. Incompatible fields may be cleared to prevent an obsolete selection from remaining in the calculation.

2.4. Cam Data

The Cam Data panel defines the follower type, rotation speed, and associated geometric parameters. For a flat follower, the base radius and calculated follower length are shown. For a roller follower, the base radius, roller radius, eccentricity, and maximum pressure angle are shown.

2.5. Motion chart and profile drawing

The upper part of the central area shows displacement over the cycle. The lower part shows the cam contour, follower, and segment boundaries. The mouse wheel changes the zoom, dragging with the left button pans the drawing, and the fit command restores the view. The playback control performs one complete revolution at the specified speed.

2.6. Result charts and tables

The right column switches between charts and tables. The data selector allows examination of cam SVAJ, follower SVAJ, contour, pitch curve, pressure angle, curvature, or combinations available for the selected follower type.

3. Project and cycle definition

3.1. Projects

The .crcm file preserves segments, curves, custom curves, settings, calculation model, compatibility criterion, motion reference, and cam parameters. When another project is started, a file is opened, or the application is closed, the application checks for unsaved changes.

3.2. Input modes and cycle closure

Angles can be entered using β or the initial and final θ limits. Displacements can be entered using ΔL or the initial and final L values. The choice changes only the editing method; the internal state maintains ordered critical points.

The sum of β may be less than 360° . In this case, the program internally creates a dwell up to 360° at the current displacement. The sum cannot exceed 360° , and all intervals must remain positive and ordered.

3.3. Motion-curve calculation models

Table 3 – Curve-selection models

Model	User control	Recommended use
Manual selection	The user selects the curve for each segment.	Directed comparison of motion laws and use of custom curves.
Automatic search	The program finds compatible sequences and presents a list of possible alternatives.	Exploration of combinations when junction conditions take priority.

When the **Possible curves only** filter is active, the selector omits curves whose start and end signatures are incompatible with adjacent segments. Disabling the filter makes the entire library available, but places responsibility for evaluating discontinuities on the user.

3.4. V, A and V, A, J criteria

The V, A criterion compares the signs of velocity and acceleration at the junctions. The V, A, J criterion also includes regular jerk. Sign matching reduces infeasible combinations, but does not replace verification of numerical values or discontinuity correction.

The **Consider impulsive events** option, available in Settings, excludes curves with infinite values, undefined quantities, or acceleration jumps at relevant points. Use this restriction when the contact model, drive system, or dynamic limits do not allow such events.

4. Closed-form and configurable curves

4.1. Closed-form curves

CrucibleCam provides the following displacement curves for selection:

- C1/C3 – Cycloid 1st half;
- C2/C4 – Cycloid 2nd half;
- C5/C6 – Complete cycloid;
- H1/H3 – Harmonic 1st half;
- H2/H4 – Harmonic 2nd half;
- H5/H6 – Complete harmonic;
- P1/P2 – 8th-degree polynomial;
- DW – Dwell;
- VC – Constant velocity;
- PA – Parabolic;
- DH – Double harmonic;
- CB1 – Cubic no. 1;
- CB2 – Cubic no. 2;
- CB3 – Cubic no. 3;
- P34 – 3-4 polynomial;
- P45 – 3-4-5 polynomial;
- P67 – 4-5-6-7 polynomial;
- P9 – 9th-degree polynomial;
- TA – Trapezoidal acceleration;
- MT – Modified trapezoidal;
- MS – Modified sinusoidal;
- MS1 – Modified sinusoidal, 1st half;
- MS2 – Modified sinusoidal, 2nd half;
- MC – Modified cycloid;
- P5 – 5th-degree polynomial;
- P11S – Symmetric 11th-degree polynomial;
- P11V – 11th-degree polynomial with midpoint velocity;
- BD – Berzak D;
- BE – Berzak E;
- G3 – Gutman 1-3 harmonic;
- F13 – Freudenstein 1-3;
- F35 – Freudenstein 1-3-5;
- WB – Weber.

4.2. GP - General polynomial

The general polynomial can operate in automatic mode or with its degree defined by the conditions. In automatic mode, CrucibleCam selects and may adjust the degree to preserve coupling. In condition-driven mode, each enabled constraint contributes to determining the degree; entered values remain fixed and take precedence over automatic coupling.

4.3. BS - B-Spline

The B-Spline is defined by its order, total number of knots, and distribution of interior knots. Endpoint knots are repeated automatically. The interface indicates whether the number of independent conditions is compatible with the number of coefficients.

4.4. Boundary and intermediate conditions for configurable curves

Conditions may apply to displacement, velocity, acceleration, and jerk at the start, end, or intermediate positions. Avoid redundant or conflicting conditions. The editor identifies singular, incomplete, overdetermined systems or systems with a residual above the tolerance.

Each configurable curve is stored as an instance linked to the segment. The edit button reopens the conditions of the selected instance. Changes to segment displacements update endpoint conditions that depend on those displacements.

5. Discontinuity correction

5.1. Diagnostics and objectives

The correction command compares the junctions and classifies differences in displacement, velocity, acceleration, and, when enabled, jerk. The diagnostics distinguish continuous, adjustable, structural, and cycle-closure cases.

Solutions can be ranked by:

- Least change to the input data;
- Lowest acceleration;
- Lowest jerk;
- Smallest estimated cam;
- Lowest maximum pressure angle when the follower is a roller follower.

The all-objectives option presents representative alternatives for each criterion.

5.2. Correction methods

Table 4 – Correction methods

Method	Quantities changed	Notes
β only	Angular spans	Keeps displacements and curves.
ΔL only	Displacement changes	Keeps angles and curves.
β and ΔL	Angles and displacements	Expands the search space.
Curve replacement	Selected curve families	Preserves the direction of motion.
Smooth transition	Segment splitting and existing curves	Does not create polynomials or splines.
Conditioned curves	Free segments converted to a polynomial or B-Spline	Keeps β and ΔL .

The **Advanced Options** allow separate limits for β and ΔL variation, impose a minimum β variation, and lock β , the curve, or displacement for each segment.

The **Enable maximum variation** option limits the changes made during discontinuity correction. When enabled, it allows separate maximum variation limits for β , as a percentage of 360°, and ΔL , as a percentage of the maximum L value; when disabled, these parameters may vary freely during the solution search.

Excessive restrictions can eliminate all solutions. Remove locks or increase the limits when the diagnostics indicate a restriction-related failure.

5.3. Preview and apply

The preview temporarily changes the main project to allow inspection of the charts and profile. Reset restores the previous state; Apply confirms the solution; Cancel ends the session without retaining the preview.

6. Cam geometry and followers

6.1. Speed and rotation direction

Speed can be displayed in rpm or rad/s. The rotation direction determines the profile orientation, relative motion, and pressure-angle sign. Changing the direction requires the associated charts and signs to be reviewed again.

6.2. Flat follower

For a flat follower, the user enters the base radius. The required length is calculated from contact throughout the cycle. The symmetry option uses the same length on the +Y and -Y sides; when disabled, the lengths are evaluated separately.

6.3. Roller follower

For a roller follower, enter the base radius, roller radius, and signed eccentricity. At zero angle, positive eccentricity places the roller center at +Y and negative eccentricity at -Y. The same convention is used for the contour, pitch curve, pressure angle, and radii of curvature.

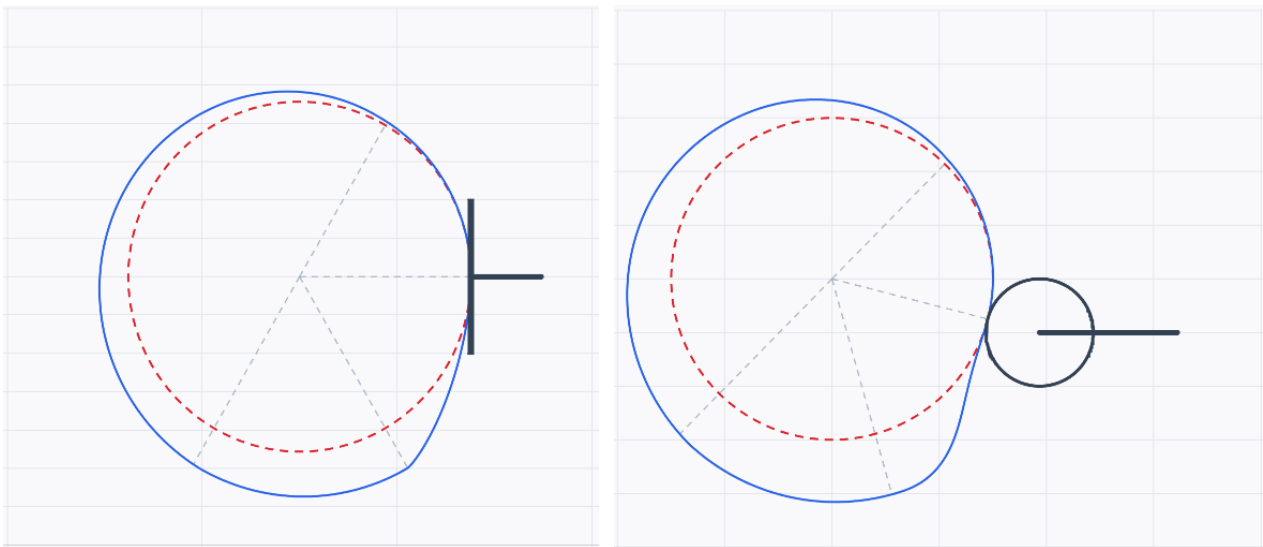


Figure 2 - Geometric parameters of flat and roller followers

6.4. Geometric recommendations

CrucibleCam calculates recommended minimum or maximum values and flags inputs that may produce sharp points, a high pressure angle, or geometric incompatibility. The drawing retains the entered value to support informed analysis; the warning does not replace the design decision.

- A base radius below the recommended minimum may produce sharp points on the profile.
- For a roller follower, the magnitude of the eccentricity must be less than the pitch radius.
- The roller radius must be positive and less than the available curvature limit.
- The maximum pressure angle must be interpreted together with the rotation direction and mechanism load.

7. Result analysis

7.1. SVAJ diagrams

The diagrams can be displayed in the cam or follower reference frame. In the cam reference frame, derivatives are taken with respect to angle. In the follower reference frame, velocity, acceleration, and jerk incorporate the specified angular speed and are presented as functions of equivalent time.

7.2. Contour and pitch curve

The cam contour is the exportable physical geometry. For a roller follower, the pitch curve represents the path of the roller center. Use the combined view to verify orientation, eccentricity, and segment boundaries.

7.3. Pressure angle and curvature

The pressure angle is shown throughout the cycle and summarized by its maximum absolute value. For a roller follower, the program calculates the radii of the pitch curve and the surface. Changes to the base radius, roller radius, eccentricity, or rotation direction require a new analysis.

7.4. Tables

The Tables tab shows the calculated motion and geometry points. Resolution controls data density.

8. Report and exports

8.1. Project report

The report contains identification, a summary, settings, cam data, segments, critical points, custom curves, and calculated extreme values. The preview window allows the content to be checked before export to PDF.

8.2. DXF export

DXF export writes the physical cam contour in the selected units. The drawing exported in DXF format represents the cam profile using straight-line segments that consecutively connect the calculated points. The file therefore does not use arcs, splines, or other curved entities to reconstruct the contour. The smoothness and geometric accuracy of the profile depend directly on the selected calculation resolution: the smaller the angular interval between points, the greater the number of segments and the closer the approximation to the theoretical profile.

Before using the file for manufacturing, verify tolerances, scale, contour closure, and the requirements of the CAD/CAM process.

8.3. XLSX export

XLSX export contains kinematic data, profile geometry, project parameters, and custom-curve definitions. The number of rows follows the solution resolution.

9. FREE and PRO features

The FREE version covers essential use, while the PRO version adds advanced features, broader deployment options, and license plans for individual users and institutions.

The PRO version unlocks the following features:

- V, A, J curve compatibility criterion;
- Solution resolution below 1° ;
- Simultaneous β and ΔL discontinuity correction;
- Curve-replacement discontinuity correction;
- Transition-based discontinuity correction;
- Discontinuity correction with conditioned curves;
- Pressure-angle optimization;
- Eccentricity adjustment during correction;
- DXF profile export;
- XLSX data export.